Amogh Ukkadgatri

Electronics & Communication Engineer · Embedded Systems · RF & Signal Processing

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SUMMARY

Electronics and communication engineer in defence R&D, on the guidance package for a small air-defence interceptor end to end: radar link-budget and detection analysis, antenna and array design with full-wave EM simulation, digital back-end architecture, and multi-board flight electronics through to fabrication-ready hardware and its measurement plan. Background in acoustic counter-UAS — mic arrays, embedded firmware and the test rigs that prove them. Suited to regulated, high-reliability defence and aerospace work.

EXPERIENCE

R&D Intern — SSS Defence
Sep 2025 – Present

Bengaluru, India (on-site) · DeepTech R&D — Air-defence interceptor guidance package, acoustic C-UAS

- Working on the guidance package for a small air-defence interceptor, scoped end to end: radar link-budget and detection-performance analysis, antenna and array design backed by full-wave electromagnetic simulation, digital back-end and processing architecture, and the multi-board flight electronics — power, interconnect and layout — through to fabrication-ready hardware and its measurement plan.
- Recruited into the acoustic counter-UAS program on the strength of a final-year sound source localization project.
- Built the microphone arrays for the first C-UAS prototype on Raspberry Pi 4 across multiple communication protocols, and adapted a triangulation-based localization approach to support the team's classical and physics-based algorithms.
- Built a complete test rig single-handedly to validate custom mic arrays over ESP32 I2S PDM capture, solving the classic ESP32's single-RX-channel limit by moving to TDM; applied DHCP, static IP and subnet masking across prototypes for Ethernet data transfer.

SELECTED PROJECTS

3D Sound Source Localization using a Mic Array — Sep 2025 – Mar 2026

Real-time audio sensing system that detects and localizes sound sources with classical DOA methods over a custom microphone array.

Real-time 3D Radar Tracking & Classification System — Sep – Nov 2024

Radar system that tracks objects in three dimensions and classifies them by their distinct signatures.

Smart Alcohol Detection & Engine Immobilization — Jan – Mar 2023

Breath-alcohol sensor wired into a vehicle ignition lock to immobilize the engine on detection.

Also: automatic rain-detection module (2024) · analog battery-level indicator, voltage-divider based (2024) · centroid finder in Python (2023). Full write-ups at amoghukkadgatri.vercel.app.

SKILLS

Languages & tools	C, Python, MATLAB, Verilog (basics), Audacity, Linux
Hardware & embedded	ESP32, Raspberry Pi, Arduino, 8051, I2S / PDM / TDM audio, sensors, op-amps, MOSFETs, transistors, soldering and firmware flashing
Systems & RF	Link-budget & detection analysis, antenna/array design, full-wave EM simulation, digital back-end architecture, multi-board layout, signal processing
Networking	DHCP, static IP, subnet masking, Ethernet data transfer

EDUCATION

B.E., Electronics & Communication — KLE Institute of Technology · **CGPA 8.84/10** — 2022 – 2026

Senior Secondary (XII), Science — Chetan PU College, Karnataka State Board · **81.00%** — 2022

Secondary (X) — Oriental Public School, State Board · **96.16%** — 2020

LEADERSHIP & ACTIVITIES

Chair, KLEIT–IEEE MTT-S Chapter (2024–25) · Vice-Chair, KLEIT–IEEE Student Branch (2023–24) · Coordinator, Technical & Cultural Fest · Class Representative (2023–24) · University Cultural Group (Theatre)